

04 “ src foundation extraction

Source: docs/research/browser_extension/Browser Torrent Extension Guide/src/ Reference browser extension **BobaLink** (WXT + TypeScript + Manifest V3). This is the foundation layer (build/config + data layer + shared libs + types). Every file below was read IN FULL. Exact values are quoted with file + line citations.

Files read

File	Lines
package.json	61
tsconfig.json	37
wxt.config.ts	169
jest.config.ts	143
playwright.config.ts	133
.eslintrc.json	56
.prettierrc	14
_locales/en/messages.json	118
.github/workflows/ci.yml	55
.github/workflows/release.yml	416
sql/schemas.sql	276
sql/migrations/001_initial.sql	267
src/types/api.ts	486
src/types/config.ts	232
src/types/torrent.ts	223
src/shared/constants.ts	403
src/shared/crypto.ts	292
src/shared/errors.ts	342
src/shared/events.ts	175
src/shared/logger.ts	167
src/shared/storage.ts	248
src/shared/utils.ts	366

1. Package metadata, dependencies, npm scripts (package.json)

- **name** bobalink, **version** 1.0.0, **type** module, **license** MIT, **author** Boba Project.
- **engines.node** >=20.0.0.
- **repo** <https://github.com/boba-project/bobalink.git> (HTTPS “ note Boba constitution mandates SSH; placeholder only).
- **description**: “Browser extension that detects torrent files and magnet links, sending them to Boba Project’s qBitTorrent dashboard”.

Dependencies (1)

Package	Version
webextension-polyfill	^0.12.0

devDependencies (15)

Package	Version
@playwright/test	^1.49.0
@types/chrome	^0.0.287
@types/jest	^29.5.14
@types/node	^22.10.0
@types/webextension-polyfill	^0.12.1
@typescript-eslint/eslint-plugin	^8.18.0
@typescript-eslint/parser	^8.18.0
@wxt-dev/auto-icons	^1.0.2
eslint	^9.17.0
jest	^29.7.0
jest-environment-jsdom	^29.7.0
prettier	^3.4.0
ts-jest	^29.2.0
ts-node	^10.9.2
typescript	^5.7.0
wxt	^0.19.0

npm scripts (all 15)

Script	Command
dev	wxt
dev:firefox	wxt -b firefox
build	wxt build
build:firefox	wxt build -b firefox
zip	wxt zip
zip:firefox	wxt zip -b firefox
compile	tsc --noEmit
lint	eslint src/ --ext .ts
lint:fix	eslint src/ --ext .ts --fix
format	prettier --write "src/**/*.ts"
format:check	prettier --check "src/**/*.ts"
test	jest --coverage
test:watch	jest --watch
test:e2e	playwright test
test:e2e:ui	playwright test --ui
postinstall	wxt prepare

Note: `package.json` references `wxt build` (chrome default) + `build:firefox`. The **release.yml** workflow references `build:chrome` / `build:firefox` scripts that **do not exist** in `package.json` (drift in the reference repo “flagged for the plan”).

2. Manifest / permissions / CSP config (`wxt.config.ts` + `_locales/en/messages.json`)

WXT generates `manifest.json` from `wxt.config.ts`. Extracted manifest config:

- **manifest_version**: 3
- **name**: `__MSG_extName__` “BobaLink”
- **description**: `__MSG_extDescription__`
- **version**: 1.0.0
- **default_locale**: en
- **minimum_chrome_version**: "109"
- **outDir**: dist

Permissions (6)

storage, alarms, notifications, activeTab, scripting, contextMenus

host_permissions (3) “the load-bearing endpoint set

- `http://localhost:7187/*` “Boba FastAPI server
- `http://localhost:7189/*` “Boba Go server
- `http://localhost:8080/*` “qBitTorrent WebUI

⚠️ Mismatch vs the real Boba port map: actual qBitTorrent WebUI proxy is **7186** (CLAUDE.md port table), merge service **7187**, boba-jackett **7189**. The extension hardcodes **8080** for qBitTorrent and assumes FastAPI on 7187 / Go on 7189. The plan must reconcile (qBittorrent proxy is 7186, not 8080).

action (toolbar popup)

- **default_popup**: `popup/index.html`
- **default_icon**: 16 “icon-16.png”, 32 “icon-32.png”
- **default_title**: `__MSG_extName__`

background (service worker)

- **service_worker**: `background.js`
- **type**: module (ES module service worker)

content_scripts (1 entry)

- **matches**: `["<all_urls>"]`
- **js**: `["content-scripts/content.js"]`
- **css**: `["content-scripts/content.css"]`
- **run_at**: `document_idle`

options_page

- options/index.html

web_accessible_resources

- resources ["assets/*"], matches ["<all_urls>"]

content_security_policy (extension_pages)

default-src 'self'; script-src 'self'; style-src 'self' 'unsafe-inline'; c

- Strict: no inline scripts (script-src 'self'); style-src allows 'unsafe-inline'; connect-src restricted to the three localhost backends; img-src allows data: URIs.

commands (keyboard shortcuts, 3)

Command	Default	Mac	Description (msg)
send-to-boba	Ctrl+Shift+B	Command+Shift+B	__MSG_cmdSendToBoba__
scan-page	Ctrl+Shift+S	Command+Shift+S	__MSG_cmdScanPage__
open-dashboard	Ctrl+Shift+D	Command+Shift+D	__MSG_cmdOpenDashboard__

WXT build options

- autoIcons.baseIconPath: src/assets/icon.svg (auto-generates icon sizes from one SVG)
- dev.port: 3000
- runner.disabled: true (no auto browser launch)

i18n messages (_locales/en/messages.json) â€” 28 keys

Key UI strings: extName=â€œBobaLinkâ€¢, extDescription, command descriptions (cmdSendToBoba/cmdScanPage/cmdOpenDashboard), popup (popupTitle=â€œBobaLink - Detected Torrentsâ€¢, noTorrentsDetected, sendToBoba, scanning, sending, sendSuccess, sendFailed), connection states (connectionOnline=â€œConnectedâ€¢, connectionOffline=â€œDisconnectedâ€¢, notConfigured=â€œNo server configuredâ€¢), options page sections (optionsTitle, serverSectionTitle, generalSectionTitle, advancedSectionTitle), buttons (addServer, testConnection, autoDiscover, saveChanges, discardChanges), type labels (magnetLink, torrentFile), context menu (contextMenuSend, contextMenuScan, contextMenuDashboard).

3. TypeScript compiler config (tsconfig.json)

Option	Value
target	ES2022

Option	Value
module	ESNext
moduleResolution	bundler
lib	["ES2022", "DOM", "DOM.Iterable"]
jsx	preserve
strict	true
noImplicitAny	true
strictNullChecks	true
strictFunctionTypes	true
strictBindCallApply	true
strictPropertyInitialization	true
noImplicitReturns	true
noFallthroughCasesInSwitch	true
noUncheckedIndexedAccess	true
exactOptionalPropertyTypes	true
noUnusedLocals	true
noUnusedParameters	true
declaration	true
declarationMap	true
sourceMap	true
esModuleInterop	true
skipLibCheck	true
forceConsistentCasingInFileNames	true
resolveJsonModule	true
isolatedModules	true
types	["chrome", "jest", "node", "webextension-polyfill"]
baseUrl	.
paths	~/* â†’ src/*, @/* â†’ src/ src/**/* .ts, tests/**/* .ts,
include	wxt.config.ts, jest.config.ts, playwright.config.ts
exclude	node_modules, dist, .wxt

Very strict TS posture (all strict flags + noUncheckedIndexedAccess + exactOptionalPropertyTypes).

ESLint (.eslintrc.json)

- parser: @typescript-eslint/parser; project: ./tsconfig.json (type-aware lint).
- extends: eslint:recommended, plugin:@typescript-eslint/recommended, ...recommended-requiring-type-checking, ...strict.
- env: browser, es2022, node, webextensions, jest.
- Notable rules: no-explicit-any **error**; no-floating-promises **error**; no-misused-promises **error**; prefer-nullish-coalescing **error**; no-console warn (allows error/warn/info); eqeqeq error (null-ignore); no-unused-vars

error with `^_ignore`. `tests/**` override turns off unsafe/any rules. Ignores `dist/`, `.wxt/`, `node_modules/`, `coverage/`.

Prettier (`.prettierrc`)

```
semi:true, trailingComma:"all", singleQuote:false, printWidth:100,
tabWidth:2, useTabs:false, bracketSpacing:true, arrowParens:"always",
endOfLine:"lf", quoteProps:"as-needed", bracketSameLine:false,
proseWrap:"preserve".
```

Jest (`jest.config.ts`)

- `preset ts-jest/presets/default-esm`, `testEnvironment: jsdom`, `ESM (extensionsToTreatAsEsm: [".ts"])`.
- `roots tests/unit+src`; `testMatch **/tests/unit/**/*test.ts, **/src/**/*test.ts`.
- `moduleNameMapper` mirrors `tsconfig` paths (`~/`, `@/`).
- **coverageThreshold global 80% (branches/functions/lines/statements)**.
- `collectCoverageFrom: src/**/*ts` excluding `*.d.ts`, `index.ts`, `assets/`, `popup/`, `options/`, `content/`, `background/` (i.e. "covers logic libs only" parsers/api/scanner/shared).
- `setupFiles tests/unit/chrome-mock.ts`; `setupFilesAfterEnv tests/unit/setup.ts`; `clearMocks+restoreMocks`; `testTimeout: 10000`; `reporters text, text-summary, lcov, html` "coverage/".

Playwright (`playwright.config.ts`)

- `testDir ./tests/e2e`; `fullyParallel: false`; `workers: 1` (extension state); `retries 2` in CI; `reporters html+list (+github in CI)`.
- `use.baseURL chrome-extension://test-id/`; `trace on-first-retry`; `screenshot only-on-failure`; `video on-first-retry`; `viewport 1280~720`.
- `launchOptions` load unpacked ext via `--load-extension=$ {EXTENSION_PATH} | ./dist`.
- `projects: chromium (channel chromium) + firefox`.
- `webServer: command: npm run build`, `url http://localhost:3000`, `timeout 120000`.
- `globalSetup/globalTeardown: ./tests/e2e/global-setup.ts / global-teardown.ts` (these files do not exist in the tree "referenced only").
- `timeout: 30000`, `expect timeout 5000`.

4. Data model "SQL schema + TypeScript types

4a. SQLite schema (`sql/schemas.sql = migrations/001_initial.sql`, **schema v1.0.0**)

Persisted via **chrome.storage.local** with an **sql.js** wrapper (header comment). 9 tables, ~25 indexes (header says "7" but actual index count is higher), 30 default site selectors, 25 default config rows. Migration file wraps everything in `BEGIN TRANSACTION;` "COMMIT"; and adds a `migration_id='001_initial'` metadata row.

Table 1 app_metadata (key/value app state) | Col | Type | Constraint |
 | key | TEXT | PRIMARY KEY | value | TEXT | NOT NULL | updated_at |
 INTEGER | NOT NULL |

Seeded keys: schema_version='1.0.0', app_name='BobaLink',
 app_version='1.0.0', installed_at, last_db_vacuum (migration also seeds
 migration_id='001_initial').

Table 2 extension_config (typed key-value settings) | Col | Type | Constraint |
 | key | TEXT | PRIMARY KEY | value | TEXT | NOT NULL | data_type |
 TEXT | CHECK IN (string,number,boolean,json,encrypted) NOT NULL |
 description | TEXT | updated_at | INTEGER | NOT NULL |

25 seeded config rows (key, value, data_type): - server.base_url=http://
 localhost:8080 (string), server.auth_method=cookie (string),
 server.api_key=(**encrypted**), `server.username`=(string),
 server.password=(**encrypted**) - `download.default\_category`=,
 download.default_save_path`, download.start_paused=false(bool), download.auto_manage=true
 -behavior.auto_scan=true, behavior.highlight_detected=true, behavior.show_notifications=true, behavior.rate_
 -security.require_https=true, security.encrypt_credentials=true, security.session_timeout=3600 (number)
 -ui.theme=auto, ui.popup_width=400, ui.popup_height=600, ui.show_tracker_list=true, ui.batch_select_defa

Table 3 discovered_torrents (per-session discovery) | Col | Type | Notes |
 | id | TEXT | PRIMARY KEY | infoHash (40-char hex) | page_url | TEXT |
 NOT NULL | page_title | TEXT | magnet_uri | TEXT | torrent_url | TEXT |
 torrent_data_b64 | TEXT | Base64 .torrent content | name | TEXT | display_name | TEXT |
 URL-decoded for UI | trackers | TEXT | JSON array | size_bytes | INTEGER | nullable |
 source_type | TEXT | CHECK IN (magnet-link,torrent-file,torrent-
 url,infohash) NOT NULL | discovery_method | TEXT | CHECK IN (link-scan,text-
 scan,site-specific>manual) DEFAULT link-scan | is_private | INTEGER |
 DEFAULT 0 | detected_at | INTEGER | NOT NULL | sent_to_boba | INTEGER | DEFAULT 0 |
 sent_at | INTEGER | boba_status | TEXT | CHECK IN
 (pending,queued,sending,added,duplicate,error,retrying) DEFAULT
 pending | boba_error | TEXT | tab_id | INTEGER | tab_group_id | INTEGER | selected |
 INTEGER | DEFAULT 1 | metadata_json | TEXT |

Table 4 download_queue (offline support) | Col | Type | Notes |
 id | INTEGER | PK AUTOINCREMENT | info_hash | TEXT | NOT NULL | magnet_uri |
 TEXT | torrent_data_b64 | TEXT | name | TEXT | category | TEXT | save_path | TEXT |
 tags | TEXT | JSON array | add_paused | INTEGER | DEFAULT 0 | skip_checking |
 INTEGER | DEFAULT 0 | sequential_download | INTEGER | DEFAULT 0 |
 first_last_piece_prio | INTEGER | DEFAULT 0 | added_at | INTEGER | NOT NULL |
 retry_count | INTEGER | DEFAULT 0 | last_retry_at | INTEGER | next_retry_at | INTEGER |
 exponential backoff | last_error | TEXT | error_code | TEXT | status | TEXT | CHECK IN
 (pending,retrying,failed_permanent,completed,cancelled) DEFAULT
 pending | server_url | TEXT | NOT NULL | server_id | INTEGER | server_config.id |

Table 5 server_config (multi-server) | Col | Type | Notes |
 id | INTEGER | PK AUTOINCREMENT | name | TEXT | NOT NULL | base_url | TEXT | NOT
 NULL UNIQUE | auth_method | TEXT | CHECK IN (none,cookie,api_key,basic) NOT
 NULL DEFAULT none | api_key_encrypted | TEXT | username | TEXT |
 password_encrypted | TEXT | is_default | INTEGER | DEFAULT 0 | is_reachable | INTEGER
 | DEFAULT 0 | last_check_at | INTEGER | last_check_result | TEXT | CHECK IN
 (success,auth_required,unreachable,error) | qbittorrent_version | TEXT |
 boba_version | TEXT | boba_fastapi_port | INTEGER | DEFAULT 7187 | boba_go_port |

INTEGER | DEFAULT **7189** || qbittorrent_port | INTEGER | DEFAULT **8080** || created_at |
 INTEGER | NOT NULL || updated_at | INTEGER | NOT NULL || notes | TEXT ||

Table 6 " send_history (immutable send log) | Col | Type | Notes | " -
 | id | INTEGER | PK AUTOINCREMENT || info_hash | TEXT | NOT NULL || name | TEXT ||
 magnet_uri | TEXT || source_type | TEXT | NOT NULL || sent_at | INTEGER | NOT NULL ||
 server_id | INTEGER | REFERENCES server_config(id) || server_url | TEXT | NOT NULL ||
 success | INTEGER | NOT NULL (1/0) || response_code | INTEGER | HTTP code ||
 error_message | TEXT || torrent_added_hash | TEXT | hash returned by qBitTorrent |

Table 7 " site_selectors (CSS selectors per site) | Col | Type | Notes | " -
 " - | id | INTEGER | PK AUTOINCREMENT || domain_pattern | TEXT | NOT NULL
 (e.g. thepiratebay.*) || selector_type | TEXT | CHECK IN (magnet-link,torrent-
 link,name,size,seeders,leechers,date,uploader) NOT NULL || css_selector |
 TEXT | NOT NULL || attribute | TEXT | e.g. href, null for text || regex_filter | TEXT |
 optional || priority | INTEGER | DEFAULT 100 (lower = higher prio) || is_enabled | INTEGER |
 DEFAULT 1 || notes | TEXT ||

30 seeded selector rows covering: thepiratebay, 1337x (magnet+torrent), nyaa.si
 (magnet+torrent), rutracker (magnet+torrent), kinozal, nnm-club, iptorrents (torrent),
 limetorrents, torrentgalaxy, glodls, torlock, eztv, yts, rarbg, demonoid, torrentfunk, yourbittorrent,
 bitsearch, bt4g, solidtorrents, knaben, snowfl, bittorrent, megapeer, badasstorrents, extratorrent.
 Most use a[href^="magnet: "] @ priority 10; direct .torrent links @ priority 20 (or 10
 for iptorrents).

Table 8 " queue_log (queue event log) | Col | Type | Notes | " -
 " - | id |
 INTEGER | PK AUTOINCREMENT || queue_item_id | INTEGER | REFERENCES
 download_queue(id) || action | TEXT | NOT NULL CHECK IN
 (enqueue,retry,success,fail,cancel,skip) || timestamp | INTEGER | NOT NULL ||
 details | TEXT | JSON || error | TEXT ||

Table 9 " statistics (daily aggregates) | Col | Type | Notes | " -
 " - | id |
 INTEGER | PK AUTOINCREMENT || stat_type | TEXT | NOT NULL CHECK IN
 (torrents_detected,torrents_sent,torrents_failed,api_calls,api_errors,queue_i
 || stat_value | INTEGER | NOT NULL DEFAULT 0 || stat_date | TEXT | NOT NULL (YYYY-
 MM-DD) || created_at | INTEGER | NOT NULL || " " | **UNIQUE(stat_type, stat_date)** |

Indexes (CREATE INDEX IF NOT EXISTS): - discovered_torrents:

idx_discovered_page_url(page_url),
 idx_discovered_status(boba_status), idx_discovered_tab(tab_id),
 idx_discovered_detected(detected_at),
 idx_discovered_selected(selected) WHERE selected=1 (partial) -
 download_queue: idx_queue_status(status),
 idx_queue_next_retry(next_retry_at), idx_queue_hash(info_hash),
 idx_queue_server(server_url) - send_history:
 idx_history_hash(info_hash), idx_history_date(sent_at),
 idx_history_server(server_id) - extension_config: idx_config_key(key) -
 server_config: idx_server_default(is_default) WHERE is_default=1 (partial),
 idx_server_url(base_url) - site_selectors:
 idx_selectors_domain(domain_pattern),
 idx_selectors_enabled(is_enabled) WHERE is_enabled=1 (partial) -
 queue_log: idx_queue_log_item(queue_item_id),
 idx_queue_log_timestamp(timestamp) - statistics:
 idx_stats_type_date(stat_type, stat_date)

Compatible with SQLite 3.38+ / sql.js. **11.4.10 note:** extension_config stores server.api_key and server.password as data_type='encrypted'; server_config has api_key_encrypted/password_encrypted columns – credentials are designed to be encrypted-at-rest (see 6 crypto).

4b. TypeScript domain types

src/types/torrent.ts

- **interface** MagnetInfo: uri:string, infohash:string (40-char hex SHA-1), displayName:string|null, trackers:readonly string[], webSeeds:readonly string[], exactLength:number|null, exactSource:string|null, keywords:readonly string[], acceptableSource:string|null, manifest:string|null, detectedAt:number, sourceElement:Element|null. (Covers magnet params xt/dn/tr/ws/xl/xs/kt/as/mt.)
- **interface** TorrentFile: url:string, filename:string, size:number|null, sameOrigin:boolean, detectedAt:number, sourceElement:Element|null.
- **interface** ParsedTorrent: infohash:string, name:string, creationDate:number|null, comment:string|null, createdBy:string|null, pieceLength:number, isPrivate:boolean, trackers:readonly string[], source:string|null, files:readonly TorrentFileInfo[], totalSize:number, numPieces:number.
- **interface** TorrentFileInfo: path:readonly string[], length:number, fullPath:string.
- **type** TorrentContentType = "magnet" | "torrent-file".
- **interface** DetectedTorrent (unified detection type): id:string, type:TorrentContentType, magnet:MagnetInfo|null, torrentFile:TorrentFile|null, displayName:string, selected:boolean (mutable), sent:boolean (mutable), sendStatus:SendStatus|null (mutable), detectedAt:number.
- **type** SendStatus = "pending" | "sending" | "success" | "error" | "queued".
- **interface** SendResult: success:boolean, torrent:DetectedTorrent, error:string|null, response:Record<string,unknown>|null, completedAt:number.
- **interface** PageScanResult: pageUrl:string, pageTitle:string, items:readonly DetectedTorrent[], magnetCount:number, torrentFileCount:number, scannedAt:number, scanDurationMs:number.

src/types/config.ts

- **type** AuthMethod = "none" | "cookie" | "api_key" | "basic" (duplicated in types/api.ts).
- **interface** ServerConfig (28 fields): id, name, url, active:boolean, authMethod:AuthMethod, username:string|null, encryptedPassword:string|null, encryptedApiKey:string|null, requestTimeout:number, verifySsl:boolean, defaultCategory:string|null, defaultSavePath:string|null, startPaused:boolean, skipHashCheck:boolean,

- ```

contentLayout:"original"|"subfolder"|"no_subfolder",
autoTMM:boolean,uploadLimit:number (KiB/s, 0=unlimited),
downloadLimit:number.

```
- **interface** ExtensionConfig: schemaVersion:number, servers:readonly ServerConfig[], activeServerId:string|null, autoScan:boolean, autoScanDelay:number, highlightTorrents:boolean, highlightStyle:"badge"|"border"|"glow", showNotifications:boolean, notificationSound:boolean, autoSend:boolean, maxHistoryItems:number, debugMode:boolean, healthCheckInterval:number (min), offlineQueue:boolean, maxOfflineQueueSize:number, showContextMenu:boolean, keyboardShortcuts:boolean, lastUpdated:number, encryptionKeyVersion:number.
  - **const** DEFAULT\_CONFIG: schemaVersion:1, servers:[], activeServerId:null, autoScan:true, autoScanDelay:2000, highlightTorrents:true, highlightStyle:"badge", showNotifications:true, notificationSound:false, autoSend:false, maxHistoryItems:100, debugMode:false, healthCheckInterval:5, offlineQueue:true, maxOfflineQueueSize:50, showContextMenu:true, keyboardShortcuts:true, lastUpdated:0, encryptionKeyVersion:1.
  - **interface** AutoDiscoveryConfig: ports:readonly number[], scanTimeout:number, scanQbittorrent:boolean, scanFastApi:boolean, scanGo:boolean.
  - **const** DEFAULT\_AUTO\_DISCOVERY: ports:[7187,7189,8080], scanTimeout:3000, all three scan flags true.
  - **interface** ConnectionTestResult: success,url,version:string|null, error:string|null, responseTimeMs, testedAt.
  - **interface** ConfigChangeEvent: key,newValue:unknown, oldValue:unknown, changedAt.

src/types/api.ts

qBitTorrent API + auth + health + Boba + queue + messaging types: - QbittorrentVersion: version:string. - QbittorrentAppPreferences: locale, save\_path, temp\_path, preallocate\_all:boolean, listen\_port, max\_active\_downloads, max\_active\_uploads, max\_active\_torrents, dl\_limit, up\_limit, web\_ui\_username (all readonly). - QbittorrentAddTorrentParams (maps /api/v2/torrents/add): urls?, torrents?:File, savepath?, cookie?, category?, tags?, skip\_checking? (âœ“trueâœ“/âœ“falseâœ“ string), paused?, root\_folder?, rename?, upLimit?:number, dlLimit?:number, autoTMM?, sequentialDownload?, firstLastPiecePrio?, contentLayout?:"Original"|"Subfolder"|"NoSubfolder", stopCondition?. - QbittorrentAddResponse: success:boolean, error?:string. - QbittorrentTorrentInfo: hash, name, magnet\_uri, size, progress (0âœ“1), dlspeed, upspeed, priority, num\_seeds, num\_leechs, ratio, eta, state, category, tags, added\_on, completion\_on, save\_path. - **Auth credential interfaces:** CookieAuthCredentials{method:"cookie",username,password}, ApiKeyAuthCredentials{method:"api\_key",apiKey}, BasicAuthCredentials{method:"basic",username,password}, NoAuthCredentials{method:"none"}; union AuthCredentials. - AuthState: method:AuthMethod, isAuthenticated:boolean, sidCookie:string|null, sidExpiresAt:number|null, basicAuthHeader:string|null,

```

apiKeyHeader:string|null, lastRefreshedAt:number|null,
consecutiveFailures:number. - type AuthMethod =
"none"|"cookie"|"api_key"|"basic". - type HealthStatus =
"healthy"|"degraded"|"unhealthy"|"unknown". - HealthCheckResult:
serverId,url,status:HealthStatus,version:string|null,
responseTimeMs,authValid:boolean,error:string|null,checkedAt. -
BobaServerInfo: name,version,features:readonly string[],
qbittorrent_connected:boolean,qbittorrent_version:string|null. -
BobaSearchResult: id,name,infohash,magnet_uri,size,seeders,leechers,
source,upload_date,category. - BobaSearchResponse: results:readonly
BobaSearchResult[],total,page,per_page. - QueueItem: id,torrent:
{infohash,magnetUri:string|null,torrentUrl:string|
null,displayName},serverId,addedAt,attempts,lastError:string|null,
lastAttemptAt:number|null,priority:"high"|"normal"|"low". -
QueueProcessResult: processed,succeeded,failed,remaining,
results:ReadonlyArray<{itemId,success,error:string|null}>. - type
MessageType (20 values): scan-page,scan-result,send-torrent,send-
result,get-detected,get-config,set-config,get-auth-state,
authenticate,health-check,health-result,queue-process,queue-
status,open-dashboard,show-notification,update-badge,torrent-
detected,selection-change. - ExtensionMessage: type:MessageType,
payload?:Record<string,unknown>,requestId?:string. -
ExtensionMessageResponse: success:boolean,
data?:Record<string,unknown>,error?:string.

```

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## 5. Shared-lib API surface (every exported fn/class signature)

### src/shared/crypto.ts â€” AES-256-GCM credential encryption (Web Crypto)

- **interface** EncryptedBundle: salt:string (b64), iv:string (b64), ciphertext:string (b64, auth tag appended), version:number.
- async function encrypt(plaintext:string, passphrase:string): Promise<EncryptedBundle> â€” generates fresh random salt(16B)+IV(12B), derives key via PBKDF2, AES-GCM encrypt (tagLength 128). Throws StorageError on empty plaintext/passphrase or failure.
- async function decrypt(bundle:EncryptedBundle, passphrase:string): Promise<string> â€” re-derives key from salt+passphrase, AES-GCM decrypt. Throws StorageError (â€œpassphrase may be incorrectâ€”) on failure.
- function generateSecurePassphrase(): string â€” 32 random bytes â†’ base64.
- function isEncrypted(value:unknown): value is EncryptedBundle â€” structural type guard.
- async function sha256(input:string): Promise<string> â€” hex SHA-256 digest.
- function simpleHash(input:string): number â€” non-crypto 32-bit hash (cache keys only).
- (private: deriveKey, generateRandomBytes, bytesToBase64, base64ToBytes.)

## src/shared/errors.ts â€” error taxonomy

- **class** BobaLinkError extends Error: fields code:string, statusCode:number|null, recoverable:boolean, cause:Error|null, context:Readonly<Record<string,unknown>>. Ctor (message, options{code?,statusCode?,recoverable?,cause?,context?}). Methods getUserMessage():string, toJSON():Record<string,unknown>. Default code BOBA\_UNKNOWN.
- **class** AuthError â€” code BOBA\_AUTH\_FAILED, statusCode default 401, recoverable true.
- **class** NetworkError â€” code BOBA\_NETWORK\_ERROR, statusCode null, recoverable true.
- **class** TorrentError â€” code BOBA\_TORRENT\_ERROR, statusCode default 400, recoverable true.
- **class** ParseError â€” code BOBA\_PARSE\_ERROR, statusCode null, recoverable **false**.
- **class** ConfigError â€” code BOBA\_CONFIG\_ERROR, recoverable true.
- **class** StorageError â€” code BOBA\_STORAGE\_ERROR, recoverable true.
- **class** RateLimitError â€” code BOBA\_RATE\_LIMITED, statusCode 429, recoverable true, extra field retryAfter:number. Ctor (message, retryAfter, options{context?}).
- **class** ServerError â€” code BOBA\_SERVER\_<status>, statusCode required, recoverable when >=500 || ===429. Ctor (message, statusCode, options).
- function isBobaLinkError(value:unknown): value is BobaLinkError.
- function normalizeError(error:unknown, context={}): BobaLinkError.

## src/shared/events.ts â€” typed event emitter

- **interface** EventMap (13 events): torrent-detected{id,type:"magnet"|"torrent-file",displayName,url}, scan-started{url,timestamp}, scan-completed{url,magnetCount,torrentFileCount,durationMs}, scan-error{url,error}, send-started{ids}, send-completed{results[{id,success}]}, send-error{id,error}, auth-state-changed{method,authenticated}, connection-status{serverId,connected,latency:number|null}, config-changed{key,newValue,oldValue}, queue-updated{size}, badge-update{count,color}, notification{title,message,type:"info"|"success"|"warning"|"error"}.
- **type** EventName = keyof EventMap; **type** EventListener<T> = (payload:EventMap[T])=>void.
- **class** TypedEventEmitter: on<T>(event,listener):()=>void (returns unsub); once<T>(event,listener):void; emit<T>(event,payload):void (try/catch per listener); off(event?):void; listenerCount(event):number; hasListeners(event):boolean.
- **const** globalEvents = new TypedEventEmitter().

## src/shared/logger.ts â€” structured logging

- type LogLevel = "debug"|"info"|"warn"|"error" (severity 0/1/2/3). Default min level info; debug when debug mode on.

- function initLogger(isDebug:boolean): void.
- function debug/info/warn/error(context:string, message:string, data?:unknown): void (error always emitted).
- function timed(context, operation): ()=>void â€” logs start, returns end fn logging duration.
- function createLogger(context:string) â†’ {debug,info,warn,error,timed} (bound). **type** Logger = ReturnType.
- function isDebugEnabled(): boolean.
- Format: [ISO-timestamp] [LEVEL] [context] message.

## src/shared/storage.ts â€” chrome.storage.local wrapper

- async function storageGet<T>(key): Promise<T|null>.
- async function storageSet<T>(key, value): Promise<void>.
- async function storageRemove(key): Promise<void>.
- async function storageGetMultiple<T>(keys:readonly string[]): Promise<ReadonlyMap<string,T|null>>.
- async function storageClearAll(): Promise<void> â€” removes only bobalink\_-prefixed keys.
- function getStorageKeys(): typeof STORAGE\_KEYS.
- function onStorageChange<T>(keys, callback): ()=>void â€” local-area only; returns unsub.
- **class** NamespacedStorage: ctor (namespace) â†’ prefix bobalink\_<namespace>; methods get<T>(key), set<T>(key,value), remove(key), getAllKeys(), clear(). All errors wrapped as StorageError.

## src/shared/utils.ts â€” utilities

- function debounce<T extends unknown[]>(fn, delay): {...args:T}:void; cancel():void}.
  - function throttle<T extends unknown[]>(fn, interval): {...args:T}:void; cancel():void}.
  - function generateId(): string â€” crypto.randomUUID or timestamp+random fallback.
  - function truncate(str, maxLength): string â€” ellipsis.
  - function escapeHtml(text): string â€” via textContent/innerHTML (XSS-safe; **DOM-dependent**, not service-worker safe).
  - function sleep(ms): Promise<void>.
  - async function retryWithBackoff<T>(fn, maxRetries=3, baseDelay=1000, maxDelay=30000): Promise<T> â€” exponential backoff + 0.3 jitter.
  - **class** TokenBucket: ctor (capacity, refillRate); consume():boolean; getAvailableTokens():number; private refill().
  - function yieldToBrowser(): Promise<void> â€” rAF or setTimeout(0).
  - async function processInChunks<T>(items, processor, chunkSize=50): Promise<void>.
  - function isValidHttpUrl(url): boolean.
  - function getDomain(url): string â€” hostname or âœ…âœ….
  - function formatBytes(bytes, decimals=2): string â€” B/KB/MB/GB/TB/PB.
  - function deepClone<T>(obj): T â€” JSON round-trip.
  - function arraysEqual<T>(a, b): boolean â€” shallow.
-

## 6. Every constant value (src/shared/constants.ts) esp. URLs/ports/timeouts/secrets

### Regex patterns

- `MAGNET_REGEX = /magnet:\?xt=urn:btih:[a-fA-F0-9]{40}[\s"'<>]*/gi`
- `MAGNET_VALIDATION_REGEX = /^magnet:\?xt=urn:btih:[a-fA-F0-9]{40}(\?:[&])[\s"'<>]*)?\$/i`
- `INFOHASH_REGEX = /xt=urn:btih:([a-fA-F0-9]{40})/i`
- `TORRENT_FILE_REGEX = /https?:\\\/[\\\/\s"'<>]+\.(?:torrent|(\?:[&])[\s"'<>]*)?)?/gi`
- `TORRENT_FILE_VALIDATION_REGEX = /^https?:\\\/\.(?:torrent|(\?:[&])[\s"'<>]*)?)?\$/i`
- `INFOHASH_HEX_REGEX = /^[a-fA-F0-9]{40}\$/`
- `INFOHASH_BASE32_REGEX = /^[A-Z2-7]{32}\$/`
- `MAGNET_DN_REGEX = /[?&]dn=([^&]*)/`
- `MAGNET_TR_REGEX = /[?&]tr=([^&]*)/g`

### Ports / URLs / endpoints (load-bearing)

- `DEFAULT_PORTS = {FAST_API:7187, GO:7189, QBITTORRENT:8080}`
- `DEFAULT_URLS = {FAST_API:"http://localhost:7187", GO:"http://localhost:7189", QBITTORRENT:"http://localhost:8080"}`
- `QBITTORRENT_ENDPOINTS:`
  - `AUTH_LOGIN: "/api/v2/auth/login", AUTH_LOGOUT: "/api/v2/auth/logout"`
  - `APP_VERSION: "/api/v2/app/version", APP_PREFERENCES: "/api/v2/app/preferences", APP_SET_PREFERENCES: "/api/v2/app/setPreferences"`
  - `TORRENTS_ADD: "/api/v2/torrents/add", TORRENTS_INFO: "/api/v2/torrents/info", TORRENTS_DELETE: "/api/v2/torrents/delete", TORRENTS_PAUSE: "/api/v2/torrents/pause", TORRENTS_RESUME: "/api/v2/torrents/resume"`
  - `TRANSFER_INFO: "/api/v2/transfer/info"`

⚠️ Port reconciliation needed: extension assumes qBitTorrent direct on **8080**; Boba™s actual proxy exposes qBitTorrent WebUI on **7186** (merge service 7187, boba-jackett 7189). FastAPI 7187 + Go 7189 match. Magnet/.torrent send goes through `/api/v2/torrents/add` on the chosen base URL.

### Timing / retry / rate limit

- `DEBOUNCE_DELAYS = {MUTATION:500, STORAGE_WRITE:250, AUTO_SCAN:1000, BADGE_UPDATE:300, NOTIFICATION:2000} (ms)`
- `REQUEST_TIMEOUTS = {DEFAULT:15000, HEALTH_CHECK:5000, AUTH:10000, ADD_TORRENT:30000, AUTO_DISCOVERY:3000} (ms)`
- `RETRY_CONFIG = {MAX_RETRIES:3, BASE_DELAY_MS:1000, MAX_DELAY_MS:30000, JITTER_FACTOR:0.3}`
- `RATE_LIMIT = {MAX_REQUESTS:10, WINDOW_MS:1000} (note: differs from DB seed behavior.rate_limit_requests=30)`

## Storage keys (STORAGE\_KEYS, all bobalink\_-prefixed)

- CONFIG: "bobalink\_config", AUTH\_STATE: "bobalink\_auth\_state", CREDENTIALS: "bobalink\_credentials", DETECTED: "bobalink\_detected", HISTORY: "bobalink\_history", QUEUE: "bobalink\_queue", HEALTH: "bobalink\_health", KEY\_MATERIAL: "bobalink\_key\_material"

## Encryption (ENCRYPTION) & §11.4.10 secret-handling scheme

- ALGORITHM: "AES-GCM", KEY\_LENGTH\_BITS: 256, IV\_LENGTH\_BYTES: 12, SALT\_LENGTH\_BYTES: 16, KDF\_ALGORITHM: "PBKDF2", KDF\_ITERATIONS: 100000, KDF\_HASH: "SHA-256", CURRENT\_KEY\_VERSION: 1.
- Scheme: passphrase → PBKDF2 (100k iters, SHA-256, 16-byte random salt) → AES-256-GCM (12-byte random IV, 128-bit auth tag). Each encrypt() is independent (fresh salt+IV). Stored bundle = {salt, iv, ciphertext, version} all base64. Aligns with Boba's AES-256-GCM master-key model (boba-jackett). The passphrase source/storage is **not defined in these foundation files** see open questions.

## UI constants

- BADGE\_COLORS = {HEALTHY: "#4CAF50", DEGRADED: "#FF9800", ERROR: "#F44336", SCANNING: "#2196F3", DETECTED: "#9C27B0", DEFAULT: "#757575"}
- ICON\_SIZES = [16, 32, 48, 128]

## Site selectors (SITE\_SELECTORS, code-level supplement to DB table)

Record<domain, selectors[]>: generic (a[href^="magnet:"], a[href\$=".torrent"], a[href\*=".torrent?"], a[href\*="download.php?id="]); plus per-domain entries for 1337x.to, thepiratebay.org/thepiratebay10.org, rarbg.to/rarbgtorrents.org, yts.mx/yts.lt, eztv.re, limetorrents.lol, torrentgalaxy.to, nyaa.si, animetosho.org, torrentz2.eu, fitgirl-repacks.site, rutracker.org, katcr.co, demonoid.is, iptorrents.com (a[href\$=".torrent"], a[href\*="download.php/"]), torrentleech.org (a[href\$=".torrent"], a[href\*="/download/"]), beyond-hd.me, passthepopcorn.me.

## Misc (EXT)

- ID: "bobalink", NAME: "BobaLink", DEFAULT\_CATEGORY: "BobaLink", MAX\_POPUP\_ITEMS: 50, MAX\_DISPLAY\_NAME\_LENGTH: 80.
-

## 7. CI flag + which build/test commands to keep as manual scripts

ðŸ©© **CONSTITUTION FLAG (Boba forbids ALL CI/CD â€” Hard Stop rule, CLAUDE.md):** the reference repo ships **two GitHub Actions workflows** that **MUST NOT** be ported:

- `.github/workflows/ci.yml` â€” jobs: lint (lint + format:check + compile), unit-test (test + coverage upload), build (npm run build + dist upload). Triggers on push/PR to main/develop.
- `.github/workflows/release.yml` â€” 6 jobs: version-bump (semver validate, CHANGELOG check, manifest version edit), build-all (matrix chrome/firefox), sign-firefox (AMO via web-ext sign + Mozilla secrets), create-release (GitHub Release), upload-chrome (Chrome Web Store publish API), release-summary. Uses secrets MOZILLA\_API\_KEY/SECRET, FIREFOX\_EXTENSION\_ID, CHROME\_CLIENT\_ID/SECRET/REFRESH\_TOKEN, CHROME\_WEBSTORE\_ID.

**Both .yml files MUST be deleted / never copied** into the Boba tree (no `.github/workflows/* .yml` allowed; CI is manual via `./ci.sh`). Extract only the **commands** worth keeping as manual scripts:

Keep as manual local commands (e.g. wired into a `ci.sh`-style script or Makefile target): - **Lint/format/compile gate:** `npm run lint` + `npm run format:check` + `npm run compile (tsc --noEmit)`. - **Unit tests + coverage:** `npm run test (jest --coverage, 80% threshold)`. - **E2E:** `npm run test:e2e (playwright test)` â€” requires built `dist/` + `globalSetup/teardown` files (absent; must be authored). - **Build:** `npm run build (chrome) / npm run build:firefox; package: npm run zip / zip:firefox`. - **Release packaging mechanics worth reusing manually** (NOT as automation): `zip dist dir, sha256sum checksum, inject VERSION/BUILD_SHA/BUILD_DATE` files into `dist`. Firefox `web-ext sign` and Chrome Web Store upload are out of scope for the local-manual model (and need real store credentials â€” handle per Â§11.4.10 if ever used, never in-repo). - The release workflowâ€™s `sed -i "s/1.0.0/$VERSION/g" src/sql/schemas.sql src/sql/migrations/001_initial.sql` shows version is also embedded in SQL seed rows â€” a manual bump script must touch `package.json` + `wxt.config.ts` manifest version + both SQL files + `app_metadata/extension_config schema_version/ app_version` rows together.

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## 8. Open questions / drift to resolve in the plan

1. **Port mismatch (critical).** Extension hardcodes qBitTorrent on **8080** (`host_permissions`, `CSP`, `DEFAULT_URLS/PORTS`, `DB` defaults). Bobaâ€™s real qBitTorrent WebUI is proxied on **7186** (CLAUDE.md port table); 7187=merge service, 7189=boba-jackett. `FastAPI(7187)/Go(7189)` match but qBitTorrent must change 8080â†’7186 across `wxt.config.ts` (`host_permissions` + `CSP connect-src`), `constants.ts`, `SQL` seeds, and `DEFAULT_AUTO_DISCOVERY.ports`.
2. **Encryption passphrase provenance undefined.** `crypto.ts` requires a passphrase for encrypt/decrypt and `STORAGE_KEYS.KEY_MATERIAL / EXTENSION/ encryptionKeyVersion` exist, but **where the passphrase comes from / how itâ€™s persisted** is not in the foundation layer (must be in background/options â€” not in scope here). Must confirm it satisfies Â§11.4.10 (no plaintext secret at rest, no secret in git). The



server.api\_key/server.password SQL seeds are empty strings (data\_type=encrypted) â€” fine.

3. **Dual storage model.** Two parallel persistence designs exist: (a) the SQL schema (sql.js over chrome.storage) with 9 tables, and (b) the STORAGE\_KEYS JSON-blob model + NamespacedStorage. The foundation does not show which is authoritative at runtime (storage.ts only implements the JSON-blob path; no sql.js wrapper file present in this layer). Plan must decide one source of truth.
4. **Type duplication.** AuthMethod defined identically in both types/api.ts and types/config.ts; ServerConfig (TS) and server\_config (SQL) diverge in field naming (url vs base\_url, encryptedPassword vs password\_encrypted, port fields only in SQL). Reconcile.
5. **Config-value mismatch.** RATE\_LIMIT.MAX\_REQUESTS=10/WINDOW\_MS=1000 (constants) vs DB seed behavior.rate\_limit\_requests=30. And autoScanDelay default 2000 (config) vs DEBOUNCE\_DELAYS.AUTO\_SCAN=1000. Decide canonical.
6. **Referenced-but-absent files** (in this layer): tests/e2e/global-setup.ts, tests/e2e/global-teardown.ts, src/assets/icon.svg, popup/options/content/background source, and the sql.js wrapper. These are downstream layers, not foundation â€” flagged so the plan doesnâ€™t assume they exist yet.
7. **Release workflow references non-existent npm scripts** (build:chrome) and non-existent src/manifest.chrome.json/manifest.firefox.json (WXT generates the manifest, there are no static manifest files). Drift in the reference repo â€” ignore for porting.
8. **Repo URL is HTTPS** (https://github.com/boba-project/bobalink.git) â€” Boba constitution mandates SSH remotes (git@â€”). Fix when integrating.
9. minimum\_chrome\_version "109" + MV3 module service worker â€” confirm target browser floor is acceptable; Firefox MV3 support (release.yml builds firefox) needs verification of background type:module SW support.